# The tadpole reabsorption lemma: normal-ordered cubic vertices in the matched bookkeeping eliminate the tadpole channel identically

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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## 1. Purpose and scope

Unit U4 identified the tadpole channel ($9M^2\hat{G}$ of $\langle V_3^2\rangle_c$) as load-bearing for any SC-SCOPE lift. This note proves the channel is absent identically in the matched bookkeeping, discharging the v1.0 residuals R-U6-1 (this writeup) and R-U6-2 (the machine cross-check). **Scope:** the matched bookkeeping of the B5 chain (per-pattern self-consistent Hartree dressing + per-pattern condensate optimization, Math426–Math437). **Not claimed:** any change to the sunset channel, any SC-SCOPE margin, any tier or gate.

## 2. The normal-ordering identity

Let $\delta$ be the Gaussian fluctuation with variance $M = \langle\delta^2\rangle$ and let $:\cdot:$ denote Wick normal-ordering with respect to it. The classical identity

$$\delta^n = \sum_{k\geq 0} \binom{n}{2k} (2k-1)!!\, M^k \,{:}\delta^{n-2k}{:}$$

(one-line proof: apply $\langle e^{\lambda\delta}\rangle = e^{\lambda^2 M/2}$ to $e^{\lambda\delta} = e^{\lambda^2 M/2}\,{:}e^{\lambda\delta}{:}$ and match powers) gives in particular $\delta^3 = 3M{:}\delta{:} + {:}\delta^3{:}$, $\delta^4 = 6M{:}\delta^2{:} + 3M^2 + {:}\delta^4{:}$, $\delta^5 = 10M{:}\delta^3{:} + 15M^2{:}\delta{:} + {:}\delta^5{:}$, $\delta^6 = 15M{:}\delta^4{:} + 45M^2{:}\delta^2{:} + 15M^3 + {:}\delta^6{:}$. All coefficients are machine-verified by exact pairing counts (`ru61_tadpole_alignment.py`, assert 2).

## 3. Exact alignment with the production bookkeeping

Write $\varphi = c + \delta$ with condensate $c = t\,\psi$, $\langle\psi^2\rangle = 1$, and expand the interaction $\frac{u}{4}\varphi^4 + \frac{v}{6}\varphi^6$ by the identity above, collecting by the normal-ordered degree $j$:

$j = 0$ **(the layer functional).** The $c^4$ coefficient is $\frac{u}{4} + \binom{6}{2}\frac{v}{6}M \cdot \langle\psi^4\rangle$-normalised $= \frac{u}{4} + \frac{5v}{2}M = \frac{u_{\text{eff}}(M)}{4}$ with $u_{\text{eff}} = u + 10vM$: EXACTLY the $0.25\,UN_4 + 2.5\,VN_4M$ line of the production engine (Math437 §1). Machine: assert 3 (equality to $10^{-15}$ at $M = M_R$).

$j = 1$ **(the source).** The coefficient of ${:}\delta{:}$ linear in $c$ is $3uM + 15vM^2$ (from $\delta^3$ and $\delta^5$): EXACTLY the gap/stationarity dressing – at the $A = 0$ gap point the production solver gives $m_R - r = 0.29952571$, reproduced by $3uM_R + 15vM_R^2$ to $5\times 10^{-7}$ (assert 4, independent solver). The $c^3$ source is $u_{\text{eff}}t^3$ and the $c^5$ source the sextic line: together the $j = 1$ collection IS $\partial\Phi/\partial t$ plus the $\hat{\kappa}$ dressing, which VANISHES at each pattern's own optimum – the matched bookkeeping's defining property.

$j = 2$. The ${:}\delta^2{:}$ collection is the per-pattern gap dressing (the production `gap_solve` line $\hat{r} = r + 3uM + 15vM^2 + 6unA^2 + 60vnA^2M + 5vN_4A^4$); resummed into $G$.

$j = 3$ **(the cubic vertex).** What remains is

$$V_3 = g_3\,{:}\delta^3{:}, \qquad g_3 = u_{\text{eff}}(M)\,c + \tfrac{10v}{3}\,c^3,$$

normal-ordered BY CONSTRUCTION – not by choice: the non-normal-ordered pieces of the cubic are precisely the $j = 1$ source already absorbed by stationarity.

## 4. The lemma (proof)

> **Lemma (tadpole reabsorption).** *In the matched bookkeeping the third-cumulant correction of any competitor pattern is* $\delta F_3 = -\frac{1}{2}\langle{:}V_3{:}^2\rangle_c = -\frac{1}{2}\sum_t |\hat{g}_3(t)|^2\, 6\,\widehat{G^3}(t)$*: the sunset channel only; the tadpole contraction* $9M^2\hat{G}$ *is absent identically.*

*Proof.* By Sec. 3 the residual cubic vertex is ${:}\delta^3{:}$. Wick's theorem for normal-ordered insertions forbids self-contractions; the brute-force enumeration of the 15 perfect matchings of the six legs of $\langle\delta^3(x)\,\delta^3(y)\rangle$ splits them as 6 cross-only ($=3!\,G(x{-}y)^3$, the sunset) $+9$ containing a self-pair (the $9M^2G$ tadpole class) – assert 1 – and exactly the 9 self-pair matchings are the ones killed by normal-ordering. Hence $\langle{:}\delta^3(x){:}{:}\delta^3(y){:}\rangle = 6\,G(x-y)^3$ and the tadpole channel does not occur. □

## 5. The two v1.0 objection cases, written out

($\beta$) **Off-optimum competitors.** The $j = 1$ source vanishes at each pattern's OWN layer optimum, which is where the chain's comparison functional evaluates it (the races optimize $(t, m)$ per pattern; Prop-A quantifies over all trials, dominating off-optimum evaluations from below). For a hypothetical off-optimum evaluation the residual linear term generates an in-layer shift whose energy gain is second-order along the shift direction – an amplitude move inside the Prop-A trial quantification, or a profile move inside the class quantification. No tadpole CONTRACTION arises in either case; only the source re-appears, and it is dominated as above.

($\gamma$) **Resonant $O(I^2)$ accounting.** The $O(c^3)$ part of $g_3$ (weight $\frac{10v}{3}t^3$) shifts the stationarity source at relative order $(10v/3)t^2/u_{\text{eff}} = 8.0\times10^{-3}$ at the endpoint ($= O(I)$, assert 8); the optimum moves by $O(I)$ and the energy by $O(I^2)\times$curvature – which is the $N_4$ resonance accounting ALREADY inside the matched second-cumulant functional (Math437 gallery). The genuinely new remainder is $O(I^4)$: $4.3\times10^{-10} < 10^{-6}$ at the endpoint (assert 8).

## 6. Self-caught correction of the v1.0 mechanism wording

v1.0 wrote: "the would-be tadpole linear source $3Mg_3 = 3Mu_{\text{eff}}F$ coincides term-by-term with the stationarity-equation source." This is IMPRECISE: with the dressed $g_3$, $3M\,u_{\text{eff}} = 3uM + 30vM^2$, whereas the exact $j = 1$ source is $3uM + 15vM^2$ – the naive contract-the-dressed-vertex route counts the double self-loop twice (symmetry factor 2), over-counting by exactly $15vM^2 = 0.58181468$ at $M_R$ (assert 6). The CORRECT mechanism is the Hermite collection of Sec. 3, in which each self-contraction is counted once with its proper factor. The lemma's conclusion is unaffected – the channel is absent either way – but the v1.0 shorthand would have failed a referee's coefficient check, so it is corrected here rather than silently rewritten.

## 7. Numerical verification

`codes/vacuum/ru61_tadpole_alignment.py` (8/8 asserts PASS, exit 0; JSON artefact `runs/260612-ru61-tadpole-alignment/result.json`): pairing split $15 = 6 + 9$ (brute force); Hermite coefficients exact (integer combinatorics + moment closure); $j = 0$ quartic $= u_{\text{eff}}/4$ to $10^{-15}$; $j = 1$ source $= m_R - r$ against the independent production gap solver; Math437 Lemma-1 boxed identity $u_{\text{eff}}(M_+) = \frac{1}{3}\sqrt{9u^2 - 60vr}$ to $10^{-12}$; the $15vM^2$ over-count quantified; the Math436/Math437

HEX argmin $\hat{r} = 0.3093733$ reproduced (clearly-labelled oracle); the endpoint remainder $4.3 \times 10^{-10}$; $u_{\text{eff}}(M_R) = 2.685$ matching the certified value.

Quantitative sanity check (mandatory): direction – striking a negative-definite channel can only raise competitor free energies (selection-favorable), fixed by combinatorics, not magnitude tuning; limit case – for $\mathcal{R}_H$ ($c = 0$) both channels vanish and the lemma is trivial; the corrected-vs-naive source discrepancy ($15vM^2$) is POSITIVE, i.e. the naive shorthand was conservative in the dangerous direction (it over-stated what stationarity absorbs), so no prior conclusion leaned on the imprecision.

## 8. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

($\alpha$) *"The Hermite identity is single-mode; the field is a continuum with pattern structure."* DISMISSED: the identity is applied to the Gaussian fluctuation field mode-wise with $M$ the cell-averaged variance – precisely the production engine's own Wick bookkeeping (Math424-AddA), whose $N_4$ lines the $j = 0/j = 1$ collections reproduce EXACTLY (asserts 3–4); the alignment is verified against the engine, not assumed.

($\beta$) *"v1.0's mechanism error suggests other coefficients may be wrong too."* VALID with mitigation: that is why v1.1 machine-verifies EVERY coefficient it uses (asserts 2–6) including the production cross-checks (independent gap solver; Lemma-1 identity; HEX argmin oracle); the one imprecision found is documented in Sec. 6, not patched silently.

($\gamma$) *"The off-optimum case (Sec. 5$\beta$) leans on Prop-A's quantification, which is $\mathcal{A}_{\text{adm}}$-scoped."* VALID with mitigation: for the $C_{\text{full}}$ statement the head package's own quantification covers the trials; the lemma itself is bookkeeping-level and pattern-agnostic – the only pattern input is WHERE the functional is evaluated, and every chain evaluation point is a per-pattern optimum by construction.

## 9. Result footer

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Result ID:               B5-BEYOND-LAYER-BOUND / tadpole-reabsorption-
                         lemma (unit U6; gate R-U6-1; task T-004)
Precise statement:       PROOF (was T3 sketch): in the matched
                         bookkeeping the residual cubic vertex is
                         normal-ordered by construction (Hermite
                         collection: j=0 reproduces the 0.25U+2.5VM
                         line, j=1 the 3uM+15vM^2 gap dressing,
                         j=3 gives g_3 = u_eff c + (10v/3)c^3); hence
                         <:d^3::d^3:> = 6G^3 and the tadpole channel
                         9M^2G is ABSENT IDENTICALLY. U4 tadpole rows
                         struck; sunset is the sole surviving
                         third-cumulant threat. SELF-CAUGHT v1.1
                         correction: the v1.0 "3M g_3" shorthand
                         over-counts the v-line by 15vM^2; the
                         Hermite route is the exact mechanism.
Scope:                   matched bookkeeping of the B5 chain; all
                         anchors. No change to the sunset channel,
                         margins, tiers, or gates.
Evidence grade:          ANALYTIC (written proof) + EXECUTED
                         (ru61_tadpole_alignment.py 8/8, exit 0).
                         Residuals R-U6-1 (this writeup) and R-U6-2
                         (the script) DISCHARGED pending operator
                         review.
Reproduction command:    python codes/vacuum/ru61_tadpole_alignment.py
Falsifier:               a matched-bookkeeping derivation exhibiting a
                         non-cancelling tadpole remainder at O(I^2) or
                         larger beyond the resonant accounting; any
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                          coefficient assert failing (8 registered).
Tier before / after:      lemma upgraded T3 sketch -> written proof;
                          tier/gate action is the OPERATOR's (B5 stays
                          T5; R-U6-1 gate flip pending review).
No-overclaim statement:   does NOT lift SC-SCOPE (the sunset endpoint
                          refinement remains the open item); does not
                          change any margin; the off-optimum case rests
                          on the chain's own quantification structure.
Next required action:     operator review -> R-U6-1 gate decision; then
                          T-031 exhaustiveness decision layer.
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